Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ]

[18CS5SE03]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER-2021

SOFTWARE QUALITY MANAGEMENT

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	Articulate the ISO standard of ATM machine with the following **i.** Functionality. **ii.** Reliability. **iii.** Portability. **iv.** Usability. **v.** Maintainability	**10**
	b.	Differentiate product quality and service quality.	**04**
2.	**a.**	Name the different QA process variations in software processes.	**04**
	b.	With a neat diagram Discuss the cause and effect of developing a fitness application by using Orthogonal Defect Classification.	**10**
3.	**a.**	Give the differences between Human Cause and Organizational Cause in Defect Prevention and Process improvement.	**07**
	b.	Describe Quality assessment and why it's needed in application development.	**07**
		OR	
4.	**a.**	Name Ten Organizational Roadblocks in achieving the organization goal.	**05**
	b.	Discuss Coverage-Based Testing & Usage-Based Testing.	**09**
5.	**a.**	Write notes on Defect Logging and Documentation.	**04**
	b.	Sketch the defect prevention cycle and explain the following **i.** Review & Inspection **ii.** Walkthrough.	**10**
		OR	
6.	**a.**	Name the critical groups which are involved in the process of defect prevention and explain it.	**03**
	b.	Illustrate the steps which are involved in Root Cause Analysis.	**11**
7.	**a.**	Describe the Basic Concepts of Reliability.	**07**
	b.	Articulate the following reliability metrics which are used to quantify the reliability of software product. **i.** MTTF (Mean Time To Failure) **ii.** MTTR (Mean Time To Repair) **iii.** MTBF (Mean Time Between Failures) **iv.** Reliability **v.** Availability	**07**
		OR	
8.	**a.**	Discuss the use of Sequential Pattern in E-commerce applications.	**07**
	b.	Show the way of progress tracking in implementation and software support tool of TBRM.	**07**